Sub-Geiger mode single-photon detector using a low-dark-current InGaAs avalanche photodiode

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Abstract *We have developed a single-photon detector that uses an InGaAs avalanche photodiode (APD) operating in sub-Geiger mode. Sub-Geiger mode operation is a technique in which an APD is operated at a bias voltage that is lower than the breakdown voltage. This mode considerably reduces afterpulse probability, and single photons that arrive randomly can be detected. In the present study, we reduced the dark count rate of the sub-Geiger mode single-photon detector by using a low-dark-current InGaAs APD. Consequently, we obtained a dark count rate (DCR) of 5.6 counts per second (cps) with single-photon detection efficiency (SPDE) of 0.2%. Our single-photon detector is comparable in DCR to that of superconducting nanowire single-photon detectors.*

Introduction

Single-photon detectors for near-infrared and telecommunication-wavelength photons are fundamental devices for quantum key distribution, quantum communication, deep-space communication, light detection and ranging (LIDAR), and time of flight (TOF). For example, single-photon detectors have so far[1] been developed using photomultiplier tubes (PMTs), avalanche photodiodes (APDs), superconducting nanowire, and superconducting transition edge sensors. Among them, InGaAs APDs are promising devices for realizing practical photon detectors because, for example, they have low bias voltages of less than 100 V, are based on commercially available technology, and can be used at Peltier cooler temperatures. The drawbacks of InGaAs APDs in Geiger mode operation, which is the conventional operation mode for detecting photons using APD, were a large dark count rate (DCR) and a high afterpulse probability[2]. Usually, InGaAs APDs have been operated in gated mode[3,4] in which InGaAs APDs are applied to periodic shot duration bias synchronized to input photon timing. In gated mode; however, InGaAs APD cannot detect photons with random input timing. Hence, it is impossible to use InGaAs APDs in gated mode for LIDER, TOF, and deep space communication with data rates of hundreds of Mbps[5].

Recently, in order to solve the above problems, low-dark-current InGaAs APDs have been used for Geiger mode. And then, to reduce the afterpulse probability, some recent studies have used techniques to reduce the generated charge within the multiplication region of an APD. Active quenching[6] rapidly decreases the bias voltage to below the breakdown voltage using a fast active quenching circuit after the APD causes the breakdown by incident photons or dark current. Using this method rather than gated mode, the generated charges can be reduced. Passive quenching[7] is a simple configuration using only a resistor connected in series to an APD. When the APD causes breakdown due to a detection event, the

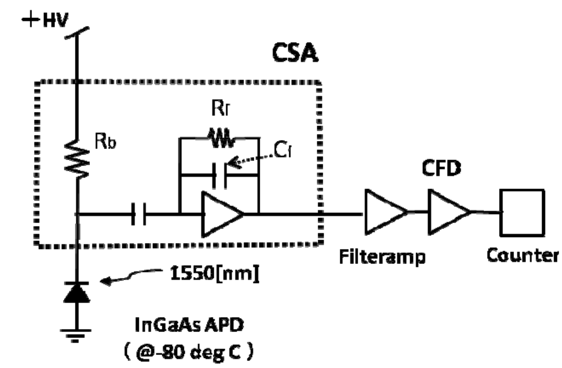

Fig. 1: Schematic diagram of sub-Geiger mode single-photon detector. CFD is an abbreviation for the constant fraction discriminator; Rf, for feedback resistance; Cf, for feedback condenser; and Rb, for bias resistance.

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generated carriers flow through the resistor causing a bias voltage drop. The time constant of the bias voltage drop depends on a series-connected resistor, APD capacitance and stray capacitance. Therefore, the charges generated by the avalanche breakdown can be reduced using optimal values of the time constant, a resistor, and excess bias voltage. Negative-feedback APD[8] is a similar approach to passive quenching. A negative-feedback APD includes a resistor for bias voltage drop. The performance of the negative-feedback APD so far is excellent: low DCR (100 cps) and low jitter (<100 ps) with conventional single-photon detection efficiency (SPDE).

We have proposed sub-Geiger mode operation as an alternative to realize free-running mode operation of InGaAs APDs[9]. Sub-Geiger mode is a technique in which the APD is operated continuously at a bias voltage below the breakdown voltage[10]. Therefore, we do not have to use a quenching circuit, because this mode never causes avalanche breakdown. This mode can considerably reduce afterpulse probability because the average number of charges generated by a detection event of 100–1000 electrons is much less than in above-Geiger mode approaches. One problem is that the number of carriers generated by APDs is considerably small, and single photons cannot be detected under such a condition using conventional amplifiers. Therefore, we must use a low-noise charge sensitive amplifier (CSA). The CSA transforms an electrical charge into a signal voltage with a high conversion gain, so it enables the detection of the faint electrical charges generated by the APD. In our previous work, we demonstrated a sub-Geiger mode single-photon detector with SPDE of 2.2% and a DCR of 7.9k counts per second (cps)[9].

In this paper, we assembled a sub-Geiger mode single-photon detector using an InGaAs APD with a dark current lower than that of the previously used APD to reduce the dark count rate. As a result, we obtained a DCR of 5.6 cps with SPDE of 0.2% using a commercially available low-noise CSA cooled to –80 ºC.

Scheme of the sub-Geiger mode single-photon detector

Figure 1 shows the scheme of the sub-Geiger mode single-photon detector. This detector consists mainly of four parts: an InGaAs APD, a CSA, a filter amplifier, and a constant fraction discriminator (CFD).

In order to reduce the dark count rate, we obtained a low-dark-current InGaAs APD (Model PGA300, Princeton Lightwave Co.). The InGaAs APD was cooled to –80 ºC in a temperature chamber (Model MC-811, ESPEC Co.) to reduce the dark current. The temperature was in the controllable range that can be achieved by a compact Peltier cooler. The dark current of this InGaAs APD as a function of the multiplication factor is indicated by the circles in Fig. 2. The fluctuation of the dark current between 10^{-15} and 10^{-17} A was due to the sensitivity limit of the measuring instrument (Model 237, Keithley Co.). The dark current of the previous InGaAs APD (Model NDL5531P, NEC Co.) as a function of the multiplication factor is shown by the triangles in Fig. 2. The dark current of the new InGaAs APD was lower by 2 orders of magnitude compared with the previous InGaAs APD.

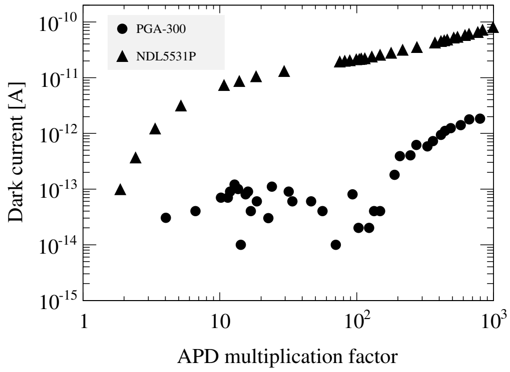

Fig. 2: APD multiplication factor and dark count rate: circles, PGA-300; triangles, NDL5531P.

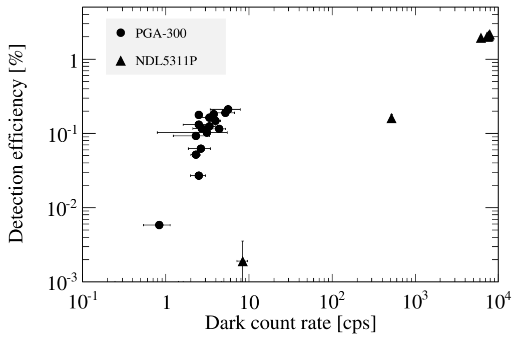

Fig. 3: Dark count rate and detection efficiency: circles, PGA-300; triangles, NDL5531P.

We connected the low-dark-current APD to a commercially available low-noise CSA (Model 581K, Clear Pulse Co.). [10] The CSA had a 0.5 pF feedback capacitor (Cf), a 5 GΩ feedback resistor (Rf), and a 1 GΩ bias resistor (Rb). The carriers generated by the APD were sent to the Cf, and the voltage value of the output became Q_{in}/Cf (Q_{in} is the input charge to the CSA). The voltage conversion gain of the CSA was 320 nV/electron. Since the decay time of the signals

Tab. 1: Comparison with other free-running single-photon detectors using InGaAs APDs

	Geiger mode			**Sub-Geiger mode**
	Active quench[6]	Passive quench[7]	Negative feedback[8]	
Detection efficiency	10%	4–5%	10%	0.2%
Dark count rate	< 2k cps	40k cps	100 cps	5.6 cps
Timing jitter	< 400 ps	< 450 ps	~59 ps	< 3.3 ns
Max count rate	31–41 kHz	4 MHz	1 MHz	> 1 MHz

from the CSA had a large time constant of 2.5 ms, we shaped the output signal by using a fast filter amplifier (Model 579, ORTEC Co.) with a shaping time constant of 10 ns. The shaped signal was discriminated by the CFD (Model 935, ORTEC Co.). The discriminated signal was counted using a pulse counter (Model SR400, Stanford Research Systems Co.) to count the dark count and photon detection events.

Single-photon detection efficiency and dark count rate

We measured the detection efficiency and the dark count rate for 1550 nm single photons. We prepared a weak laser pulse with an average of 0.1 photons per pulse generated by heavily attenuating a picosecond pulse laser at a wavelength of 1550 nm. The repetition ratio of the pulse laser was set to 20 kHz. The circles in Fig. 3 show the measured single-photon detection efficiency as a function of the dark count rate. The average multiplication factor of APD was limited to 600 at a bias voltage 59.205 V due to saturation of our amplifier systems. The maximum SPDE obtained was 0.2% at a DCR of 5.6 cps. The ratio of the single-photon detection efficiency to dark count rate (SPDE/DCR) was 3.6×10^{-2}. The results from the previous InGaAs APD (Model NDL5531P, NEC Co.) are shown by the triangles in Fig. 3, and SPDE/DCR was estimated to be 2.8×10^{-4}. Thus, we improved the performance of the sub-Geiger mode single-photon detector in terms of SPDE/DCR by adopting a new InGaAs APD.

Conclusion

We assembled a sub-Geiger mode single-photon detector using an InGaAs APD with a dark current lower than that of previously used APD and improved the performance. As a result, we obtained the DCR of 5.6 cps at SPDE = 0.2%. This value is as low as that of a superconducting single-photon detector[1]. The low detection efficiency can be improved by optimizing readout amplifier by reducing the amplifier noise and solving the saturation effect. Additionally, we measured the maximum counting rate, timing jitter, and afterpulse probability. In preliminary results, the maximum counting rate was up to 10^6 cps, timing jitter was less than 3.3 ns, and afterpulse probability was less than 4%. The timing jitter is about 10 times greater than that in Geiger mode operation. We might improve the timing jitter by decreasing the rise time of the CSA by using a faster CSA (for example, the A250 made by Amptek Inc.).

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